

EE 66
Fall 2026

Signals, Dynamics, and Information

Homework 3

This homework is due Friday, September 18, at 23:59.

Submission Format

Your homework submission should consist of a single PDF file that contains all of your answers (any handwritten answers should be scanned).

1. Some (Ortho)Normal Basis Basics

The set defined by the span of a set of vectors is always a valid subspace for a given vector space. This means that spans are also valid vector spaces. For each of the following vector spaces:

- (i) Give the dimension of the space.
- (ii) Provide an *orthonormal* basis for the vector space.

$$(a) \mathbb{O} = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} -3 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \right\}$$

Solution: The first vector is the only one with a nonzero third entry, so any linear combination of the three cannot be zero if the scalar coefficient of the first vector is nonzero. Additionally, the second vector is linearly independent of the third vector, since the two are not scalar multiples of one another. Therefore, there is no linear combination of the three vectors that makes the zero vector for non-zero coefficients. By definition, this means that all three vectors are linearly independent.

Therefore, since $\mathbb{O}$ is the span of three linearly independent vectors, **it has dimension 3**. Moreover, this implies that $\mathbb{O} = \mathbb{R}^3$ since $\mathbb{O}$ is a subspace in $\mathbb{R}^3$. Therefore, one simple orthonormal basis of $\mathbb{O}$ is the standard basis:

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$(b) \mathbb{P} = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ -4 \\ 0 \end{bmatrix} \right\}$$

Solution: With this set of vectors, we can see that the two vectors are linearly independent because they are not scalar multiples of one another. Therefore, the space $\mathbb{P}$ is the span of two linearly independent vectors, so **it has dimension 2**.

Additionally, we can see that neither vector has a non-zero third entry, so we can describe our space $\mathbb{P}$ as:

$$\left\{ \mathbf{r} = \begin{bmatrix} \alpha \\ \beta \\ 0 \end{bmatrix} \mid \forall \alpha, \beta \in \mathbb{R} \right\}$$

Following this observation, we can conclude that a valid orthonormal basis for this space consists of the first two standard basis vectors:

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$$(c) \mathbb{S} = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Solution: In the case of these vectors, we observe that the inner product is as follows:

$$\left\langle \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\rangle = [1 \quad 1 \quad 1] \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = 1 \cdot (-1) + 1 \cdot 0 + 1 \cdot 1 = -1 + 0 + 1 = 0$$

Because their inner product is 0, our two vectors are orthogonal to one another. Consequently, they are linearly independent of one another, so our space $\mathbb{S}$, which is the span of two linearly independent vectors, is of **dimension 2**.

To convert our orthogonal vectors into *orthonormal* vectors, we normalize them by dividing them by their own magnitude. This operation preserves their direction (and thus span), but ensures that the resulting vectors will be unit length.

$$\left\| \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3} \quad \text{and} \quad \left\| \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\| = \sqrt{(-1)^2 + 0^2 + 1^2} = \sqrt{2}$$

$$\mathbf{s}_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix} \quad \text{and} \quad \mathbf{s}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

Therefore an orthonormal basis that spans $\mathbb{S}$ is:

$$\left\{ \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} \right\}$$

$$(d) \mathbb{T} = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ -3 \end{bmatrix} \right\}$$

Solution: In the case of these vectors, we observe that the fourth vector is simply a scalar multiple of the third vector. Therefore, it is linearly dependent with the other three vectors. The remaining three vectors, however, do not share any non-zero entries, making them linearly independent of one another. Therefore, $\mathbb{T}$ can be characterized by the span of the first three vectors.

As the span of three linearly independent vectors, $\mathbb{T}$ has **dimension 3**.

Note that the inner product of any pair of these three vectors is zero which means that they form an orthogonal basis of the space $\mathbb{T}$. Thus, similarly to the previous subpart, we can generate an *orthonormal* basis of $\mathbb{T}$ by simply normalizing the three orthogonal vectors of interest:

$$\left\| \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \right\| = \sqrt{1^2 + 1^2 + 0^2 + 0^2 + 0^2} = \sqrt{2}$$

$$\left\| \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} \right\| = \sqrt{0^2 + 0^2 + 1^2 + (-1)^2 + 0^2} = \sqrt{2}$$

$$\left\| \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 2 \end{bmatrix} \right\| = \sqrt{0^2 + 0^2 + 0^2 + 0^2 + 2^2} = 2$$

$$\mathbf{t}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{t}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}$$

$$\mathbf{t}_3 = \frac{1}{2} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Therefore, an orthonormal basis of $\mathbb{T}$ is:

$$\left\{ \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

2. Span Proofs

- (a) Given some set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, show the following:

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\} = \text{span}\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_1 - \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$$

In other words, we can simultaneously replace $\mathbf{v}_1$ and $\mathbf{v}_2$ with their sum and difference without changing the span.

To show this, you have to prove the two following statements:

- If a vector $\mathbf{q}$ belongs in $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, then it must also belong in $\text{span}\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_1 - \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$.
- If a vector $\mathbf{r}$ belongs in $\text{span}\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_1 - \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$, then it must also belong in $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$.

In summary, you have to prove the problem statement from both directions.

Hint: Can you write $\mathbf{v}_1$ and $\mathbf{v}_2$ each as a linear combination of $(\mathbf{v}_1 + \mathbf{v}_2)$ and $(\mathbf{v}_1 - \mathbf{v}_2)$?

Solution:

Proof writing techniques:

This is the most common approach when showing that two sets are equal, $A = B$

- Show that an arbitrary element of set A is in B . $a \in A \implies a \in B$ which is the same as $A \subseteq B$.
- Show that an arbitrary element of set B is in A . $b \in B \implies b \in A$ which is the same as $B \subseteq A$.

If both statements are true, then $A = B$ must necessarily be true.

Forward Direction:

We can write $\mathbf{v}_1$ and $\mathbf{v}_2$ as linear combinations of $(\mathbf{v}_1 + \mathbf{v}_2)$ and $(\mathbf{v}_1 - \mathbf{v}_2)$:

$$\mathbf{v}_1 = \frac{1}{2}[(\mathbf{v}_1 + \mathbf{v}_2) + (\mathbf{v}_1 - \mathbf{v}_2)], \quad \mathbf{v}_2 = \frac{1}{2}[(\mathbf{v}_1 + \mathbf{v}_2) - (\mathbf{v}_1 - \mathbf{v}_2)].$$

So $\mathbf{v}_1, \mathbf{v}_2 \in \text{span}\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_1 - \mathbf{v}_2\}$. Since $\mathbf{v}_3, \dots, \mathbf{v}_n$ appear in both sets, any linear combination of $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ can be rewritten as a linear combination of $\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_1 - \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$.

Backward Direction:

Now, we must show the other direction. Suppose we have some arbitrary $\mathbf{r} \in \text{span}\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_1 - \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$. For some scalars b_i :

$$\mathbf{r} = b_1(\mathbf{v}_1 + \mathbf{v}_2) + b_2(\mathbf{v}_1 - \mathbf{v}_2) + b_3\mathbf{v}_3 + \dots + b_n\mathbf{v}_n.$$

Expanding:

$$\mathbf{r} = (b_1 + b_2)\mathbf{v}_1 + (b_1 - b_2)\mathbf{v}_2 + b_3\mathbf{v}_3 + \dots + b_n\mathbf{v}_n,$$

which is a linear combination of $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, so $\mathbf{r} \in \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$. Combining this with the earlier result, the spans are the same.

- (b) Consider the span of the set $\{\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{u}\}$. Suppose $\mathbf{u}$ is in the span of $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. Then, show that any vector $\mathbf{r}$ in $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{u}\}$ is in $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$.

Hint: Start by applying the definition of span for the $\mathbf{u}$ and $\mathbf{r}$ vectors.

Solution:

Given:

From the first sentence of the question, by definition of span, we know that any vector $\mathbf{r}$ in $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{u}\}$ can be written $\mathbf{r} = m\mathbf{u} + a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n$. Using the summation symbol, we can also write $\mathbf{r} = m\mathbf{u} + \sum_{k=1}^n a_k\mathbf{v}_k$.

From the second sentence of the question, since $\mathbf{u}$ is in the span of $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$, we can write $\mathbf{u} = b_1\mathbf{v}_1 + b_2\mathbf{v}_2 + \dots + b_n\mathbf{v}_n$ or $\sum_{k=1}^n b_k\mathbf{v}_k$.

Proof:

Now that we have written out the two assumptions from the problem statement, let's put them together by substituting the expression for $\mathbf{u}$ into the previous expression.

$$\begin{aligned}\mathbf{r} &= m\mathbf{u} + \sum_{k=1}^n a_k\mathbf{v}_k \\ \mathbf{r} &= m\left(\sum_{k=1}^n b_k\mathbf{v}_k\right) + \sum_{k=1}^n a_k\mathbf{v}_k\end{aligned}$$

Finally, gathering up coefficients, we get:

$$\mathbf{r} = \sum_{k=1}^n (mb_k + a_k)\mathbf{v}_k,$$

where $(mb_k + a_k)$ are just real numbers, so this arbitrary vector $\mathbf{r}$ is also in $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ since it is a linear combination of $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$.

Intuitively, this means that $\mathbf{u}$ is redundant so we can safely remove it without reducing our span. More rigorously, adding a vector, $\mathbf{u}$, which is linearly dependent to a set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$, to the span of those vectors will not change the span of the vectors.

Another valid solution is showing that $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\} = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{u}\}$, which clearly implies that $\mathbf{r} \in \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$.

- (c) Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a linearly independent set. Define a new set of vectors by pairing up neighbors and wrapping around:

$$\mathbf{u}_i = \mathbf{v}_i + \mathbf{v}_{i+1} \text{ for } i = 1, \dots, n-1, \quad \mathbf{u}_n = \mathbf{v}_n + \mathbf{v}_1.$$

(i) Suppose n is **odd**. Prove that $\text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\} = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$.

(ii) Suppose n is **even**. Prove that $\text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\} \neq \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$.

Hint: If every $\mathbf{v}$ is a linear combination of all the $\mathbf{u}_i$'s, what can you say about $\text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ and $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$? What about if every $\mathbf{u}$ is a linear combination of the $\mathbf{v}_i$'s?

Hint: Try to solve for $\mathbf{v}_1$ in terms of the $\mathbf{u}_i$'s by substitution: use $\mathbf{u}_1 = \mathbf{v}_1 + \mathbf{v}_2$ to write $\mathbf{v}_2$ in terms of $\mathbf{v}_1$, then use $\mathbf{u}_2$ to write $\mathbf{v}_3$ in terms of $\mathbf{v}_1$, and so on around the cycle. You should notice a pattern. Extrapolate to write an equation for, say, $\mathbf{v}_1$ in terms of all the $\mathbf{u}_i$'s. Can you write a similar equation for any $\mathbf{v}_k$? What do you notice about this equation? Does it change depending on the parity of n ?

Solution:

Each $\mathbf{u}_i$ is by definition a linear combination of $\mathbf{v}_i$'s, so $\text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\} \subseteq \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. The question is whether the reverse also holds.

Following the hint, we substitute around the cycle. From $\mathbf{u}_i = \mathbf{v}_i + \mathbf{v}_{i+1}$, we can solve $\mathbf{v}_{i+1} = \mathbf{u}_i - \mathbf{v}_i$ and chain these together:

$$\begin{aligned}\mathbf{v}_2 &= \mathbf{u}_1 - \mathbf{v}_1 \\ \mathbf{v}_3 &= \mathbf{u}_2 - \mathbf{v}_2 = \mathbf{u}_2 - \mathbf{u}_1 + \mathbf{v}_1 \\ \mathbf{v}_4 &= \mathbf{u}_3 - \mathbf{v}_3 = \mathbf{u}_3 - \mathbf{u}_2 + \mathbf{u}_1 - \mathbf{v}_1 \\ &\vdots\end{aligned}$$

At each step, the signs on the $\mathbf{u}$'s alternate, and $\mathbf{v}_1$ survives with coefficient $(-1)^{k-1}$ in $\mathbf{v}_k$. In general:

$$\mathbf{v}_k = \mathbf{u}_{k-1} - \mathbf{u}_{k-2} + \cdots + (-1)^k \mathbf{u}_1 + (-1)^{k-1} \mathbf{v}_1.$$

Substituting $\mathbf{v}_n$ into the last equation $\mathbf{u}_n = \mathbf{v}_n + \mathbf{v}_1$ gives:

$$\mathbf{u}_n = (\text{alternating sum of } \mathbf{u}_1, \dots, \mathbf{u}_{n-1}) + ((-1)^{n-1} + 1) \mathbf{v}_1.$$

Case (i), n odd: $(-1)^{n-1} = +1$, so the coefficient of $\mathbf{v}_1$ is 2. Solving:

$$\mathbf{v}_1 = \frac{1}{2}(\mathbf{u}_1 - \mathbf{u}_2 + \mathbf{u}_3 - \cdots + \mathbf{u}_n).$$

By cyclically relabeling (applying the same substitution starting at $\mathbf{u}_k$ instead of $\mathbf{u}_1$), we can recover every $\mathbf{v}_k$ as a linear combination of the $\mathbf{u}_i$'s. This shows $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\} \subseteq \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$, and combined with the inclusion noted at the start, the two spans are equal.

Case (ii), n even: $(-1)^{n-1} = -1$, so the coefficient of $\mathbf{v}_1$ is 0. The $\mathbf{v}_1$ drops out entirely, and what remains is:

$$\mathbf{u}_1 - \mathbf{u}_2 + \mathbf{u}_3 - \cdots - \mathbf{u}_n = \mathbf{0},$$

a nontrivial linear dependence among $\mathbf{u}_1, \dots, \mathbf{u}_n$ (the coefficients ± 1 are all nonzero). So $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ is linearly dependent and spans a subspace of dimension at most $n-1$. But $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is linearly independent and spans an n -dimensional subspace. Since $n-1 < n$, we cannot have $\text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\} = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$.

3. Signal Spaces: Bounded and Finite-Energy Signals

Consider a complex-valued DT signal $x : \mathbb{Z} \rightarrow \mathbb{C}$. We can define ℓ_1 , ℓ_2 , and ℓ_∞ norms for these signals:

$$\|x\|_1 = \sum_{n=-\infty}^{\infty} |x[n]|, \quad \|x\|_2 = \left(\sum_{n=-\infty}^{\infty} |x[n]|^2 \right)^{1/2}, \quad \|x\|_\infty = \sup_{n \in \mathbb{Z}} |x[n]|.$$

where $|\cdot|$ is the complex magnitude: $|\alpha + i\beta| = \sqrt{\alpha^2 + \beta^2}$ and $\sup_{n \in \mathbb{Z}}$ denotes the least upper bound on the signal (this is often functionally the same as $\max_{n \in \mathbb{Z}}$).

As an engineer you'll encounter lots of signals, and you'll probably ask the following questions about them: *Will it ever blow up?* and *Will it go on forever?* Consider the feedback whine of a microphone, a resounding "yes!" to both questions! To help answer these mathematically, we can evaluate the following conditions:

- **Absolutely summable:** $\ell^1(\mathbb{Z}) = \{x : \|x\|_1 < \infty\}$. The magnitudes $|x[n]|$ add up to a finite number, so $x[n]$ must tend to 0 as $|n| \rightarrow \infty$. This is the strongest condition.
- **Finite energy:** $\ell^2(\mathbb{Z}) = \{x : \|x\|_2 < \infty\}$. The squared magnitudes $|x[n]|^2$ add up to a finite number, so $x[n]$ must also tend to 0 as $|n| \rightarrow \infty$. This is a weaker condition, but perhaps more physical.
- **Bounded:** $\ell^\infty(\mathbb{Z}) = \{x : \|x\|_\infty < \infty\}$. For some finite M and all n , $|x[n]| \leq M$. Here, $x[n]$ doesn't need to tend to 0. This is the weakest condition, but it is often the most commonly used!

Useful facts:

- $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$ for $|r| < 1$.
 - $\sum_{k=1}^{\infty} \frac{1}{k^s}$ is finite if and only if $s > 1$. In particular, $\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$.
 - $u[n]$ is the unit step: $u[n] = 1$ for $n \geq 0$, and 0 otherwise.
- (a) For each of the following signals, find $\|x\|_1$, $\|x\|_2$, and $\|x\|_\infty$ (write ∞ if a norm diverges), and thus determine the strictest condition it satisfies. Also, verify that $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$.
- $x[n] = \left(-\frac{1}{2}\right)^n u[n]$
 - $x[n] = \frac{1}{n+1} u[n]$
 - $x[n] = e^{i\omega_0 n}$ for a fixed real ω_0
 - $x[n] = n u[n]$

Solution:

i. **Absolutely summable.**

$$\|x\|_1 = 2, \quad \|x\|_2 = \frac{2}{\sqrt{3}}, \quad \|x\|_\infty = 1.$$

Norms ignore the alternating sign, so $|x[n]| = \left(\frac{1}{2}\right)^n$. The geometric series gives $\|x\|_1 = \sum_{n \geq 0} \left(\frac{1}{2}\right)^n = 2$ and $\|x\|_2^2 = \sum_{n \geq 0} \left(\frac{1}{4}\right)^n = \frac{4}{3}$. The peak is $|x[0]| = 1$.

ii. **Finite energy.**

$$\|x\|_1 = \infty, \quad \|x\|_2 = \frac{\pi}{\sqrt{6}}, \quad \|x\|_\infty = 1.$$

Substituting $k = n + 1$ gives $\|x\|_1 = \sum_{k \geq 1} \frac{1}{k}$, which diverges, and $\|x\|_2^2 = \sum_{k \geq 1} \frac{1}{k^2} = \frac{\pi^2}{6}$. The terms $\frac{1}{k}$ decay too slowly to sum, but their squares $\frac{1}{k^2}$ decay fast enough.

iii. **Bounded.**

$$\|x\|_1 = \infty, \quad \|x\|_2 = \infty, \quad \|x\|_\infty = 1.$$

$|x[n]| = 1$ for all n , so the signal never exceeds 1 but never decays.

iv. **None.**

$$\|x\|_1 = \infty, \quad \|x\|_2 = \infty, \quad \|x\|_\infty = \infty.$$

$x[n]$ grows without bound.

Each signal satisfies $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$. This holds for every DT signal, so $\ell^1(\mathbb{Z}) \subseteq \ell^2(\mathbb{Z}) \subseteq \ell^\infty(\mathbb{Z})$.

Exam tip: based on the above string of inequalities, you might be able to save some time by checking the peak first: if $\|x\|_\infty = \infty$, x is in none of these subspaces. Then check the sum: if $\|x\|_1 < \infty$, x is in all three. You only need to compute $\|x\|_2$ if x is bounded but $\|x\|_1 = \infty$!

- (b) Let $y[n] = x[-n]$ (time reversal) and $z[n] = x[n - k]$ (a shift by an integer k). Show that $\|y\|_p = \|z\|_p = \|x\|_p$ for $p = 1, 2, \infty$.

Solution: Time reversal and shifting only relabel the indices: as n runs over all of $\mathbb{Z}$, so do $-n$ and $n - k$. So y and z take exactly the same values as x , just at different indices. For example, substituting $m = -n$,

$$\|y\|_1 = \sum_n |x[-n]| = \sum_m |x[m]| = \|x\|_1.$$

The same substitution works for $\|\cdot\|_2$ and $\|\cdot\|_\infty$, and $m = n - k$ handles z . Hence time reversal and shifts preserve boundedness and finite energy.

- (c) Show that the set of bounded signals, $\ell^\infty(\mathbb{Z}) = \{x : \|x\|_\infty < \infty\}$, is a subspace of the set of all DT signals. Consider also the set of *unbounded* signals, $\mathcal{U} = \{x : \|x\|_\infty = \infty\}$. Is *this* a subspace? Why or why not?

Hint: Just as vectors add and scale entry by entry, DT signals add and scale sample by sample. It might be helpful to bound $|x[n] + y[n]|$ by a constant that does not depend on n .

Hint: to show a set is not a subspace, it is enough to exhibit one subspace axiom that fails.

Solution:

- **Zero:** the zero signal has $\|0\|_\infty = 0 < \infty$.
- **Scaling:** for every n , $|\alpha x[n]| = |\alpha| |x[n]| \leq |\alpha| \|x\|_\infty$, so $\|\alpha x\|_\infty < \infty$.
- **Addition:** for every n , the triangle inequality gives $|x[n] + y[n]| \leq |x[n]| + |y[n]| \leq \|x\|_\infty + \|y\|_\infty$. This bound does not depend on n , so $\|x + y\|_\infty \leq \|x\|_\infty + \|y\|_\infty < \infty$.

The key to the addition step is the intermediate bound $|x[n] + y[n]|$: the triangle inequality bounds each sample of $x + y$ by it, and replacing each term with its peak gives one constant that works for every n . The triangle inequality for complex numbers is

$$|a + b| \leq |a| + |b| \quad \text{for all } a, b \in \mathbb{C}.$$

The set $\mathcal{U}$ of unbounded signals is *not* a subspace. The zero signal has $\|0\|_\infty = 0 < \infty$, so $0 \notin \mathcal{U}$. A subspace must contain the zero vector.

(Alternatively, via addition: the signal $x[n] = nu[n]$ from part (a)(iv) is unbounded, and so is $-x$, but $x + (-x) = 0$ is bounded. Two elements of $\mathcal{U}$ can sum to something outside $\mathcal{U}$.)

4. Periodicity of Complex Exponentials

A discrete-time signal $x : \mathbb{Z} \rightarrow \mathbb{C}$ is p -periodic if it repeats every p timesteps. That is,

$$x[n+p] = x[n] \quad \forall n \in \mathbb{Z}. \quad (1)$$

Note that Equation 1 holds for all multiples of periods (like $p, 2p, 3p, \dots$). The smallest $p \in \mathbb{Z}^+$ for which Equation 1 holds is called the fundamental period of $x[n]$: the shortest length after which the signal's pattern first repeats.

Let $a : \mathbb{Z} \rightarrow \mathbb{R}$ be a discrete-time signal with fundamental period p_a and $b : \mathbb{Z} \rightarrow \mathbb{R}$ be a discrete-time signal with fundamental period p_b , where p_a and p_b are coprime positive integers (that is, p_a and p_b have no common factors other than 1).

- (a) Let $c[n] = 100a[n]$. Show that the fundamental period of c is p_a .

Solution:

$$c[n+p_a] = 100a[n+p_a] \quad \forall n \in \mathbb{Z}$$

$$c[n+p_a] = 100a[n]$$

$$c[n+p_a] = c[n]$$

Thus, scaling a signal does not change its fundamental period.

- (b) Let $d[n] = e^{ip_a n}$. Is d periodic? If so, what is its fundamental period?

Solution: The signal is not periodic. If a positive integer N were a period, then

$$e^{ip_a(n+N)} = e^{ip_a n} \implies e^{ip_a N} = 1.$$

This would require $p_a N = 2\pi m$ for some $m \in \mathbb{Z}$. Since $p_a N$ is a nonzero integer and π is irrational, this is impossible.

- (c) Now let $g[n] = e^{2\pi i n}$. Is g periodic? If so, what is its fundamental period?

Solution: For every integer n ,

$$g[n] = e^{2\pi i n} = 1.$$

Thus, g is a constant signal, so its fundamental period is 1.

- (d) Let

$$f[n] = \frac{1}{2} \exp\left(\frac{2\pi i}{p_a} n\right).$$

Is f periodic? If so, what is its fundamental period?

Solution: We have

$$f[n+p] = f[n] \iff \exp\left(\frac{2\pi i}{p_a} p\right) = 1 \iff p \text{ is a multiple of } p_a.$$

Therefore, f is periodic with fundamental period p_a . The nonzero factor $\frac{1}{2}$ does not change the period, as shown in part 1.

- (e) Let $h[n] = f[n] + b[n]$. Show that h is $(p_a p_b)$ -periodic.

Solution: Since $p_a p_b$ is a multiple of both p_a and p_b ,

$$\begin{aligned} h[n+p_a p_b] &= f[n+p_a p_b] + b[n+p_a p_b] \\ &= f[n] + b[n] \\ &= h[n]. \end{aligned}$$

Thus, $p_a p_b$ is a period of h .

- (f) Now instead suppose that p_a is an integer multiple of p_b , so $p_a = kp_b$ for some $k \in \mathbb{Z}^+$. Is h still periodic? Give a period of h .

Solution: Since $p_a = kp_b$, the integer p_a is a period of both f and b . Hence,

$$h[n + p_a] = f[n + p_a] + b[n + p_a] = f[n] + b[n] = h[n].$$

Thus, h is periodic and p_a is a period.

- (g) Based on the previous two questions, what can you say about a period of the sum of two periodic signals? Be sure to address the role of coprimality.

Solution: If two signals have fundamental periods p_a and p_b , then

$$\text{lcm}(p_a, p_b)$$

is always a period of their sum because it is a multiple of both individual periods. If p_a and p_b are coprime, then

$$\text{lcm}(p_a, p_b) = p_a p_b.$$

If they are not coprime, the least common multiple is smaller than $p_a p_b$; for example, when $p_a = kp_b$, it is p_a .

- (h) Now consider

$$q[n] = \cos^2\left(\frac{2\pi}{p_a}n\right).$$

Assume that p_a is even. Rewrite q using complex exponentials and the inverse Euler formula, and show that $\frac{p_a}{2}$ is a period of q .

Solution: By the inverse Euler formula,

$$\begin{aligned} q[n] &= \left(\frac{e^{\frac{2\pi i}{p_a}n} + e^{-\frac{2\pi i}{p_a}n}}{2} \right)^2 \\ &= \frac{1}{4} \left(e^{\frac{4\pi i}{p_a}n} + 2 + e^{-\frac{4\pi i}{p_a}n} \right) \\ &= \frac{1}{4} e^{\frac{4\pi i}{p_a}n} + \frac{1}{2} + \frac{1}{4} e^{-\frac{4\pi i}{p_a}n}. \end{aligned}$$

The constant $\frac{1}{2}$ comes from the two cross terms: $e^{\frac{2\pi i}{p_a}n} e^{-\frac{2\pi i}{p_a}n} = 1$.

Since p_a is even, $\frac{p_a}{2}$ is an integer. Shifting n by $\frac{p_a}{2}$ gives

$$\begin{aligned} q\left[n + \frac{p_a}{2}\right] &= \frac{1}{4} e^{\frac{4\pi i}{p_a}\left(n + \frac{p_a}{2}\right)} + \frac{1}{2} + \frac{1}{4} e^{-\frac{4\pi i}{p_a}\left(n + \frac{p_a}{2}\right)} \\ &= \frac{1}{4} e^{\frac{4\pi i}{p_a}n} e^{2\pi i} + \frac{1}{2} + \frac{1}{4} e^{-\frac{4\pi i}{p_a}n} e^{-2\pi i} \\ &= q[n]. \end{aligned}$$

Thus, $\frac{p_a}{2}$ is a period of q . Intuitively, after $\frac{p_a}{2}$ steps the original cosine changes sign, but squaring removes that sign change.

- (i) Finally, consider

$$y[n] = \cos\left(\frac{2\pi}{p_a}n\right) \cos\left(\frac{2\pi}{p_b}n\right),$$

where p_a and p_b are coprime. Rewrite y using complex exponentials and show that it is $(p_a p_b)$ -periodic. What does this example suggest about a period of the product of two periodic signals?

Solution: Using the inverse Euler formula for each cosine,

$$\begin{aligned} y[n] &= \left(\frac{e^{\frac{2\pi i}{p_a} n} + e^{-\frac{2\pi i}{p_a} n}}{2} \right) \left(\frac{e^{\frac{2\pi i}{p_b} n} + e^{-\frac{2\pi i}{p_b} n}}{2} \right) \\ &= \frac{1}{4} \left(e^{\frac{2\pi i}{p_a} n} e^{\frac{2\pi i}{p_b} n} + e^{\frac{2\pi i}{p_a} n} e^{-\frac{2\pi i}{p_b} n} + e^{-\frac{2\pi i}{p_a} n} e^{\frac{2\pi i}{p_b} n} + e^{-\frac{2\pi i}{p_a} n} e^{-\frac{2\pi i}{p_b} n} \right). \end{aligned}$$

We now shift n by $p_a p_b$. Every exponential containing p_a gains a factor

$$e^{\pm \frac{2\pi i}{p_a} (p_a p_b)} = e^{\pm 2\pi i p_b} = 1,$$

and every exponential containing p_b gains a factor

$$e^{\pm \frac{2\pi i}{p_b} (p_a p_b)} = e^{\pm 2\pi i p_a} = 1.$$

Thus, every term in the expansion repeats after $p_a p_b$ steps, so

$$y[n + p_a p_b] = y[n].$$

Therefore, y is $(p_a p_b)$ -periodic.

The product of two discrete-time periodic signals is always periodic. In particular, if p_a and p_b are periods of the two signals, then

$$\text{lcm}(p_a, p_b)$$

is a period of their product, though it need not be the fundamental period.

5. Homework Reflection

Doing homework is an **investment in your education**, and the course staff wants to hear about your experience with this problem set! After finishing the rest of the homework, fill out these brief reflections on what you got out of your time. We'll read this and take your answers into account as we write future assignments, in this semester and future ones.

Note that we are *genuinely curious* to read what you have to write. We expect *brief but thoughtful* responses. **Hence, this part will be weighed significantly in your homework completion grade.**

- (a) Did you work on this homework alone or in a group? Did this work well for you? *We encourage you to attempt each problem on your own first, and to work with a group when you're stuck.*

Solution:

I worked alone because my study group was out of town.

I prefer to work alone because a group can be very distracting for me.

I worked with my friends Alice and Bob. It's nice to be able to bounce ideas off one another and fill in each other's gaps. Explaining my ideas to someone else is hard, but it really makes sure I understand them well!

I met some new friends Carol and Danny at homework party, and we were able to finish most of the homework together with occasional help from a tutor. I reviewed my answers on my own to make sure I understood everything.

- (b) What resources did you use to solve the homework problems? How did you use these resources, and how did they help you? Feel free to mention LLM tools, if you've used any.

Solution:

I didn't remember some of the definitions I learned in linear algebra, so I consulted my old notes and reviewed concepts as I needed to while solving the problems

I went to office hours to work on the homework, and whenever I got stuck, I asked one of the tutors to provide some hints or review concepts I didn't remember.

I tried all the problems on my own, and then asked an LLM to critique my approach. I got 90% of the way there on my own, but this interaction enriched my understanding of the material!

I reviewed my lecture notes before starting the homework; and whenever I got stuck, I scoured the discussion worksheets for similar problems and used the solutions to guide my thought process.

I asked some questions on Ed whenever I was confused so that I could get the most fitting answer to my question, in the context of EE66. I also answered some questions because this is a good way to practice the stuff I know well.

I couldn't even understand what problem 2 was asking, so I asked an AI tool to review all the relevant prerequisite knowledge and explain the problem statement in a different way before trying to tackle it on my own.

- (c) What was the most fun / interesting problem on this homework? What was the least fun / un-interesting problem?

Solution:

I found question 3 very intriguing because this is a totally different perspective on DTFS to what we learned in lecture. Question 2 was pretty boring and mechanical, because we were just practicing the synthesis equation, but I suppose it is important to be able to do these more-basic things too.

- (d) Did you feel prepared to tackle these problems? Were there any problems that seemed completely disjoint from what you've seen in class? Was there anything covered in class that you weren't able practice in this assignment?

Solution: I did feel prepared! But, question 5 seemed to me to be out of place because it's been two weeks since we covered least squares. We talked about SVD and its application for PCA in class and discussion this week, and I didn't really understand any of it, so it would have been nice to have some problems which gave me an opportunity to practice those concepts.

(e) Is there anything else you would like to share?

Solution:

Johnny, who is a tutor, was SUPER DUPER helpful in office hours. He explained stuff in such a way that I still felt like I was discovering the answer to the problems by myself, which was awesome.

I'm feeling very overwhelmed by this class and the upcoming midterm is really stressing me out. These homeworks are really hard and I wish there were more resources available, which I could use to understand them deeply.